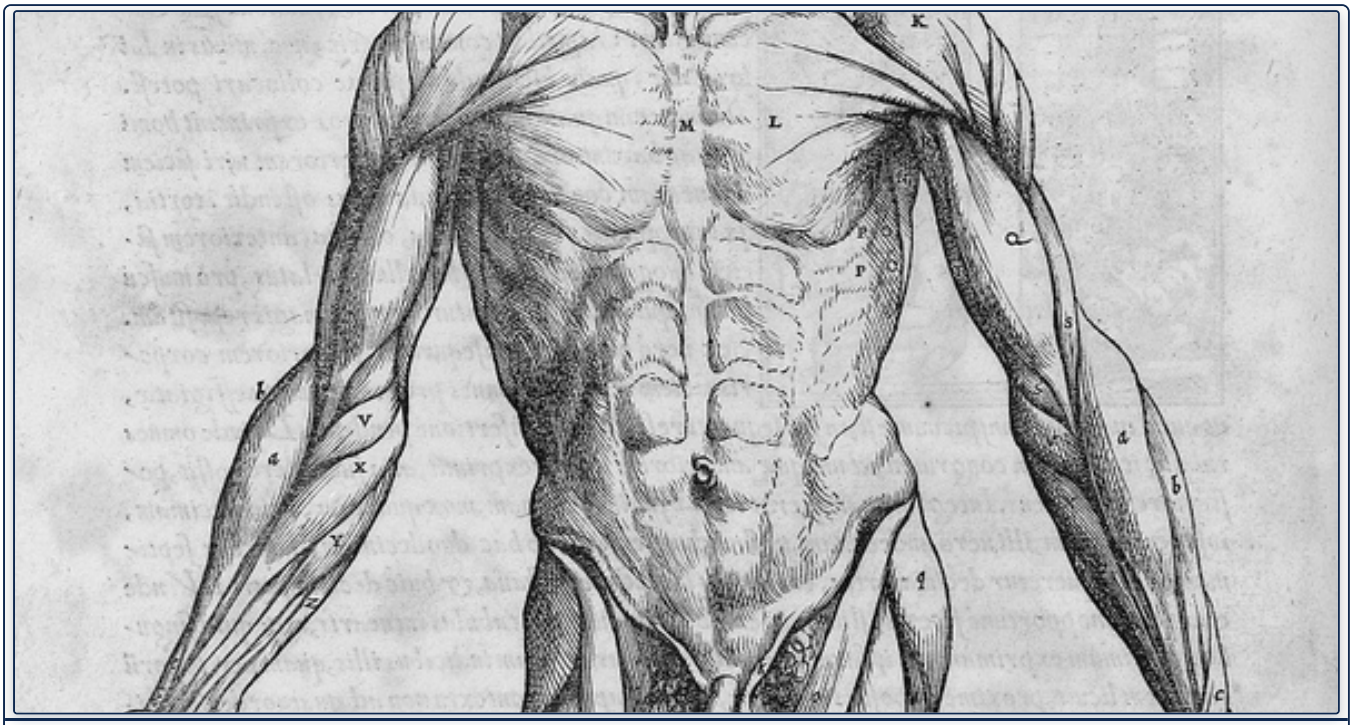
KEY TOPIC 2 • PUPIL WORKBOOK

# The Medical Renaissance in Britain, c1500–c1700

*Humanism, Anatomy (Vesalius), Circulation (Harvey), Observation (Sydenham) & The Great Plague*



**Primary Visual Evidence:** Title Page of Andreas Vesalius, *De Humani Corporis Fabrica* (1543) CONTEMPORARY PRINT ARCHIVE

PUPIL NAME: _____ TARGET: _____ CLASS: _____

| KEY TOPIC 2 SPECIFICATION FRAMEWORK (1HI0/11) | LESSON DONE | QUIZ [10] |
|---|---|---|
| **Key Topic 2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society**<br>*How did movable type (c1440), humanism, and the Royal Society (1660) challenge ancient dogma?* | ☐ | ☐ |
| **Key Topic 2.2: Thomas Sydenham & The Art of Bedside Observation**<br>*Why was Sydenham called the 'English Hippocrates', and how did Observationes Medicae classify illness?* | ☐ | ☐ |
| **Key Topic 2.3: Andreas Vesalius & The Anatomical Revolution**<br>*How did De Humani Corporis Fabrica (1543) correct over 200 of Galen's anatomical errors?* | ☐ | ☐ |
| **Key Topic 2.4: William Harvey & The Circulation of the Blood**<br>*How did De Motu Cordis (1628) prove blood circulates through the heart as a mechanical pump?* | ☐ | ☐ |
| **Key Topic 2.5: Continuity in Treatment & The Great Plague of London (1665)**<br>*Why did red crosses, quarantine, and humoral remedies in 1665 show so much continuity with 1348?* | ☐ | ☐ |

**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### c. 1440–1476 • The Printing Press: Gutenberg & William Caxton

**Key Topic 2.1**

Johannes Gutenberg invents movable metal type in Mainz; William Caxton establishes England's first press at Westminster in 1476. Medical texts can now be mass-produced with identical diagrams, shattering the Catholic Church's clerical monopoly on manuscript copying.

### 1543 • Andreas Vesalius Publishes De Humani Corporis Fabrica

**Key Topic 2.3**

At the University of Padua, professor Andreas Vesalius carries out human dissections personally. The Fabric of the Human Body features precise woodcuts disproving over 200 of Galen's errors, proving the human jawbone is one piece and the breastbone has three segments.

### 1628 • William Harvey Publishes De Motu Cordis

**Key Topic 2.4**

Royal physician William Harvey publishes On the Motion of the Heart. Through dissection of cold-blooded reptiles and mathematical calculations, he proves that the heart acts as a mechanical pump recirculating blood continuously through one-way pocket valves.

### 1660 • Foundation of the Royal Society & Nullius in Verba

**Key Topic 2.1**

The Royal Society is founded at Gresham College, London, receiving a royal charter from Charles II in 1662. Adopting the motto Nullius in Verba ('Take nobody's word for it'), it launches Philosophical Transactions in 1665 to publish and peer-review scientific experiments.



*Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two.*

**Living Timeline Dual-Coding Protocol:** Draw visual symbols for clinical bedside observation, London pest houses, microscope animalcules, and Sydenham's disease classification across the timeline below.

### 1661 • Marcello Malpighi Observes Capillaries Under the Microscope

**Key Topic 2.4**

Using an early compound microscope, Italian biologist Marcello Malpighi observes blood passing through microscopic capillaries connecting arteries to veins in a frog's lungs. This provides the definitive optical proof for William Harvey's circulatory model.

### 1665 • The Great Plague of London: Red Crosses & Pest Houses

**Key Topic 2.5**

Bubonic plague strikes London, killing approximately 100,000 people (20% of the city). The Lord Mayor orders watchmen to lock infected families inside houses marked with red crosses and 'Lord have mercy upon us'. Over 40,000 stray dogs and cats are slaughtered.

### 1665 • Robert Hooke Publishes Micrographia

**Key Topic 2.1**

Robert Hooke presents detailed microscopic drawings of fleas, cork cells, and fly eyes to the Royal Society. While microscopes prove the existence of an invisible world, lenses are too weak to reveal bacteria or germs, leaving medical treatments largely unchanged.

### 1676 • Thomas Sydenham Publishes Observationes Medicae

**Key Topic 2.2**

Known as the 'English Hippocrates', Thomas Sydenham urges doctors to base treatments on bedside clinical observation. In Observationes Medicae, he classifies illnesses into distinct species, distinguishes measles from scarlet fever, and popularises cinchona bark (quinine) for malaria.

# How did new technologies and scientific societies challenge traditional medical authority?

**Core Specification Focus:** Medical Renaissance: Humanism; the printing press (c1440); the Royal Society (1660); microscopes and scientific observation.

## 1 c1440–1476 • Why did printing break the clerical monopoly on medical learning? [1.1]

| | |
|---|---|
| • Casting lead type in Gutenberg's German workshop | |
| • Setting up William Caxton's Westminster press | |
| • Printing hundreds of identical illustrated medical books | |

## 2 c1500 • How did humanism inspire physicians to challenge Galen? [1.2]

| | |
|---|---|
| • Translating original Greek manuscripts directly | |
| • Spotting errors made by medieval monk copyists | |
| • Urging independent questioning of ancient authority | |

## 3 1660 • Why was experimental observation central to the Royal Society? [1.3]

| | |
|---|---|
| • Meeting at Gresham College in London | |
| • Receiving royal charter from King Charles II | |
| • Adopting Latin motto "Nullius in Verba" | |

## 4 1665 • How did scientific journals speed up the exchange of medical knowledge? [1.4]

| | |
|---|---|
| • Publishing Henry Oldenburg's monthly scientific journal | |
| • Distributing experimental reports across Europe | |
| • Debating laboratory findings among rival scientists | |

## 5 1665–1670s • Why did 17th-century microscopes fail to transform medical treatment? [1.5]

| | |
|---|---|
| • Grinding curved glass lenses in brass cylinders | |
| • Sketching flea legs in Hooke's Micrographia | |
| • Observing tiny swimming creatures in pond water | |



*Gutenberg gave the world movable type; the Royal Society gave the world peer review. Neither cured a cold, but both destroyed ignorance.*

## Question 4: Explain Why [12 marks • 15 mins]



**Explain why new ideas in medicine were able to spread more rapidly in the period c1500–c1700. [12 marks]**

**Stimulus:** **The invention of the printing press (c1440)** | **The Royal Society (1660)** *(You must also use information of your own)*

**ACCESSIBLE WRITING FRAMES:**

*One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:**

Gutenberg press (c1440) • William Caxton (1476) • clerical monopoly • vernacular translations • Royal Society (1660) • Royal Charter (1662) • Nullius in Verba • Philosophical Transactions (1665) • peer review • animalcules • Robert Hooke (1665)

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.1). In the drawing box, sketch Caxton's printing press beside the Royal Society's coat of arms (motto: Nullius in Verba). Annotate how printing removed the Church's control over book production. **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [ / 4 ] • 5 MINS

**Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]**

*Focus on persistent beliefs in miasma, supernatural forces, and humoral imbalance despite new anatomical knowledge.*



*Explain the mechanism: the Royal Society's journal (Philosophical Transactions) allowed European scientists to confirm and repeat experiments.*

# Why was Thomas Sydenham known as the 'English Hippocrates'?

**Core Specification Focus:** Thomas Sydenham ('The English Hippocrates'); bedside observation; classification of diseases (Observationes Medicae, 1676); treatments.

## 1 1660s • Why did Sydenham believe bedside observation surpassed textbook learning? [2.1]

- **Visiting sick London patients in private bedrooms**
- **Ignoring ancient scholastic university lectures**
- **Recording clinical symptoms in detailed notebooks**

## 2 1676 • How did classifying diseases challenge traditional individual humoral theory? [2.2]

- **Classifying distinct illnesses like modern botanists**
- **Distinguishing measles from scarlet fever rashes**
- **Publishing authoritative textbook used for two centuries**

## 3 c1670s • Why was the success of Peruvian bark a mortal blow to humoral balance? [2.3]

- **Importing bitter Peruvian tree bark to London**
- **Dosing marshland ague fevers with powdered bark**
- **Proving specific remedies cured specific diseases**

## 4 c1670s • Why did Sydenham's cool treatment drastically cut smallpox deaths? [2.4]

- **Throwing off heavy baronial wool blankets**
- **Opening window shutters to let in cold wind**
- **Giving cool drinks instead of scalding wines**

## 5 1680s • Why did medical treatments remain mostly medieval despite Sydenham? [2.5]

- **Continuing to believe miasma caused epidemics**
- **Lacking laboratory tools to identify bacteria**
- **Leaving ordinary patient treatment largely unchanged**

*Sydenham told medical students: "Go to the bedside, there alone can you learn disease." Textbooks are great, but patients actually have the symptoms.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why Thomas Sydenham was an important figure in the development of medicine in the seventeenth century. [12 marks]**

**Stimulus:** **Observationes Medicae (1676)** | **Treatments for smallpox and malaria (cinchona bark)** *(You must also use information of your own)*

**ACCESSIBLE WRITING FRAMES:**

*One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:**

Thomas Sydenham • English Hippocrates • bedside clinical observation • Observationes Medicae (1676) • species of disease • scarlet fever vs. measles • Cinchona bark (quinine) • cool therapy for smallpox • laudanum

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.2). In the drawing box, sketch Sydenham taking clinical bedside notes beside a patient with cool fresh air flowing through an open window. Annotate why his classification of diseases was revolutionary. **[ ] Done**



*Cool therapy for smallpox: opening windows and throwing off blankets. Radical in 1670, entirely sensible today.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the treatment of disease in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]**

*Focus on Sydenham's specific treatments (e.g. Peruvian bark for malaria) versus universal humoral purging.*

*Explain the impact: by proving cinchona bark cured only ague (malaria), Sydenham proved specific remedies could target specific diseases.*

# How did Andreas Vesalius revolutionize the study of human anatomy?

**Core Specification Focus:** Andreas Vesalius; The Fabric of the Human Body (De Humani Corporis Fabrica, 1543); human dissection; correcting Galen's errors.

## 1 1537 • Why was a professor dissecting human bodies revolutionary in 1537? [3.1]

- **Climbing down from high lecturer chairs**
- **Picking up steel dissection scalpels directly**
- **Opening executed criminal bodies on wooden slabs**

## 2 1540 • How did Vesalius prove Galen had made anatomical mistakes? [3.2]

- **Showing human jawbones consist of one piece**
- **Proving human breastbones have three segments**
- **Demonstrating Galen dissected pigs and Barbary apes**

## 3 1543 • Why was The Fabrica considered a masterpiece of Renaissance science? [3.3]

- **Printing seven massive anatomical volumes in Basel**
- **Engraving precise woodcut plates of human muscles**
- **Dedication presented to Holy Roman Emperor Charles V**

## 4 1545–1553 • How did English translations spread Vesalius's ideas to barber-surgeons? [3.4]

- **Copying Vesalius's woodcuts onto English copper plates**
- **Translating Latin descriptions into everyday English**
- **Selling anatomy manuals to London barber-surgeons**

## 5 c1560 • Why did Vesalius's anatomical discoveries fail to save sick patients immediately? [3.5]

- **Facing angry attacks from conservative university professors**
- **Resigning academic post to serve royal courts**
- **Acknowledging anatomy did not yet cure living sickness**



*Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two. Awkward for Galenists!*

## Question 4: Explain Why [12 marks • 15 mins]



**Explain why Andreas Vesalius made significant breakthroughs in the understanding of human anatomy in the sixteenth century. [12 marks]**

**Stimulus:** **De Humani Corporis Fabrica (1543)** | **Dissection of human corpses rather than animals** *(You must also use information of your own)*

**ACCESSIBLE WRITING FRAMES:**

*One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:**

Andreas Vesalius • Padua University • human dissection • De Humani Corporis Fabrica (1543) • anatomical woodcuts • jawbone (mandible) • breastbone segments • Barbary apes vs. humans • Thomas Geminus

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.3). In the drawing box, sketch Vesalius standing over an anatomical dissection table holding a scalpel. Annotate two specific errors of Galen that he disproved. **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which methods of anatomical training in the Renaissance (c1500–c1700) were similar to anatomical training in the nineteenth century. [4 marks]**

*Focus on direct physical dissection of human bodies as the foundation of surgical education.*



*Explain the limitation: while Vesalius proved Galen wrong on over 200 anatomical points, his discoveries did not immediately cure a single disease.*

# How did William Harvey prove the circulation of the blood, and why was it resisted?

**Core Specification Focus:** William Harvey; circulation of the blood; De Motu Cordis (1628); the heart as a mechanical pump; veins and one-way valves; limitations.

## 1 c1615 • Why did Harvey study cold-blooded animals to understand the human heart? [4.1]

| | |
|---|---|
| • Observing slow heartbeats in live cold frogs | |
| • Dissecting human corpses at the Royal College | |
| • Mapping one-way pocket valves inside veins | |

## 2 1620s • How did mathematical calculations prove Galen's theory impossible? [4.2]

| | |
|---|---|
| • Comparing the beating heart to mechanical water pumps | |
| • Calculating liver could not produce gallons of blood daily | |
| • Proving the same blood circulates continuously | |

## 3 1628 • Why did Harvey publish his research abroad rather than in England? [4.3]

| | |
|---|---|
| • Publishing Latin treatise on heart motion in Frankfurt | |
| • Demonstrating tied forearm vein experiments | |
| • Dedicating medical breakthrough to King Charles I | |

## 4 1630s–1650s • Why was Harvey's discovery fiercely rejected by established doctors? [4.4]

| | |
|---|---|
| • Being mocked as a madman by rival doctors | |
| • Losing wealthy private London medical patients | |
| • Waiting fifty years for Oxford university to teach it | |

## 5 1661 • Why was the discovery of capillaries the final proof Harvey was right? [4.5]

| | |
|---|---|
| • Focusing microscope lenses on live frog lungs | |
| • Discovering microscopic capillary vessels connecting arteries | |
| • Confirming Harvey's circulatory model beyond all doubt | |

*Galen thought the liver made fresh blood like a factory; Harvey calculated you would need to produce 250 litres a day. Mathematics: the ultimate Galen-slayer.*

## Question 4: Explain Why [12 marks • 15 mins]



**Explain why William Harvey's discovery of the circulation of the blood was a major breakthrough in medicine in the seventeenth century. [12 marks]**

**Stimulus:** **De Motu Cordis (1628)** | **Mechanical pump calculations and vein experiments** *(You must also use information of your own)*

**ACCESSIBLE WRITING FRAMES:**

*One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:**

William Harvey • De Motu Cordis (1628) • circulation of the blood • mechanical pump • one-way pocket valves • tied-arm ligature experiment • quantitative calculation • Marcello Malpighi (1661) • capillaries

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch Harvey's tied-arm ligature experiment showing blood pooling at the valves. Annotate his mathematical calculation proving Galen's liver-blood theory was impossible. **[ ] Done**



*Harvey tied a bandage around an arm and watched the veins swell. Simple experiment, monumental consequence for human physiology.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the human body in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]**

*Contrast Harvey's dynamic mechanical circulation model with Galen's belief that the liver manufactured fresh blood consumed by organs.*

# Why did treatments during the Great Plague of 1665 show so much continuity with 1348?

**Core Specification Focus:** Continuity in treatment and prevention; the Great Plague of London (1665); comparison with the Black Death; plague orders; quarantine.

## 1 Spring 1665 • Why did the rich flee London while the poor suffered catastrophic mortality? [5.1]

| | |
|---|---|
| • Spreading through poor crowded London parishes | |
| • Fleeing town by carriage: King Charles II and physicians | |
| • Leaving poor weavers trapped behind city gates | |

## 2 Summer 1665 • Why did people still use 14th-century remedies in 1665? [5.2]

| | |
|---|---|
| • Blaming planetary alignments and wrath of God | |
| • Sniffing dried sponges soaked in vinegar | |
| • Strapping plucked live pigeons to hot buboes | |

## 3 1665 • How did local government action in 1665 show greater organization than 1348? [5.3]

| | |
|---|---|
| • Painting red crosses on barred front doors | |
| • Posting armed watchmen to enforce quarantine | |
| • Building wooden pest houses outside city walls | |

## 4 1665 • Why was slaughtering cats and dogs a tragic medical mistake? [5.4]

| | |
|---|---|
| • Killing forty thousand domestic dogs and cats | |
| • Burning barrels of black tar at street crossings | |
| • Smoking tobacco pipes to ward off foul miasma | |

## 5 Winter 1665 • How devastating was the demographic impact of the 1665 epidemic? [5.5]

| | |
|---|---|
| • Ringing bell carts through silent dark lanes | |
| • Calling "Bring out your dead" into empty alleys | |
| • Burying one hundred thousand Londoners in mass trenches | |



*1665: They killed 40,000 dogs and cats to stop the plague. The real culprits—the rat fleas—enjoyed a predator-free paradise.*

## Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

**'Approaches to dealing with the Great Plague in 1665 were completely different from approaches to the Black Death in 1348.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]**

**Stimulus:** **Mayor's plague orders and quarantine (red crosses)** | **Beliefs in miasma, astrology, and supernatural causes** *(You must also use information of your own)*

**ACCESSIBLE WRITING FRAMES:**

*One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:**

Great Plague of London (1665) • Lord Mayor's orders • searchers of the dead • red crosses • "Lord have mercy upon us" • quarantine • pest houses • slaughter of 40,000 dogs and cats • smoking tar barrels • dead carts

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch a London front door marked with a red cross and guarded by a watchman. Annotate two continuities and two differences between 1348 and 1665. **[ ] Done**

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [ / 4 ] • 5 MINS

**Explain one way in which government responses to the Great Plague of London (1665) were different from government responses to the Black Death (1348). [4 marks]**

*Contrast systematic quarantine, red crosses, and pest houses in 1665 with rudimentary 1349 street cleaning.*

## KEY TOPIC 2 NOTES & EXAMINATION STRATEGY

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*Notes: Record key causal links, essay plans, and teacher feedback for this unit.*

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

| ENQUIRY | ASSESSMENT COMPONENT FOCUS | PAGE | SCORE | TEACHER COMMENT & NEXT STEPS |
|---|---|---|---|---|
| KT2.1 | **Q4:** Explain Why New Medical Ideas Spread More Rapidly c1500–c1700 | p. 5–6 | / 12 | |
| | **Q3:** Explain One Similarity (Ideas on Cause: Renaissance vs Medieval) | p. 6 | / 4 | |
| KT2.2 | **Q4:** Explain Why Thomas Sydenham Was an Important Figure | p. 8–9 | / 12 | |
| | **Q3:** Explain One Difference (Treatments: Renaissance vs Medieval) | p. 9 | / 4 | |
| KT2.3 | **Q4:** Explain Why Andreas Vesalius Made Anatomical Breakthroughs | p. 11–12 | / 12 | |
| | **Q3:** Explain One Similarity (Anatomical Training: Renaissance vs 19th C) | p. 12 | / 4 | |
| KT2.4 | **Q4:** Explain Why William Harvey's Circulation Discovery Was Major Breakthrough | p. 14–15 | / 12 | |
| | **Q3:** Explain One Difference (Ideas About Human Body: Renaissance vs Medieval) | p. 15 | / 4 | |
| KT2.5 | **Q5/6:** Evaluative Essay: Great Plague (1665) vs Black Death (1348) | p. 17–18 | / 20 | |
| | **Q3:** Explain One Difference (Government Responses: 1665 vs 1348) | p. 18 | / 4 | |
| **TOTAL** | **Key Topic 2 Cumulative Examination Assessment Portfolio (10 Tasks)** | p. 4–18 | / 88 | **Overall Grade:** _____ **Teacher Target:** _____ |

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

| KT2.1 | KT2.2 | KT2.3 | KT2.4 | KT2.5 |
|---|---|---|---|---|
| New Spirit | Thomas Sydenham | Andreas Vesalius | William Harvey | Great Plague |
| LESSON HUB | LESSON HUB | LESSON HUB | LESSON HUB | LESSON HUB |
| PRACTICE QUIZ | PRACTICE QUIZ | PRACTICE QUIZ | PRACTICE QUIZ | PRACTICE QUIZ |
| Score: [ / 10 ] | Score: [ / 10 ] | Score: [ / 10 ] | Score: [ / 10 ] | Score: [ / 10 ] |



*Nullius in Verba: "Take nobody's word for it." Scanning these quiz QR codes will prove whether your memory holds up under exam pressure.*